

TOWARD PART-AWARE CHORAL TRANSCRIPTION WITH SINGING VOICE ASSIGNMENT

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ABSTRACT

Automatic Music Transcription (AMT) has progressed substantially for single-instrument music, but choral singing remains challenging. Recent choral transcription systems can produce note-level MIDI transcriptions, yet they typically collapse all voices into a single track, losing the SATB structure needed for rehearsal, education, and score reconstruction. To address this gap, we introduce **Part-aware Choral Transcription (PawCT)**, which converts choral recordings into four note-level symbolic tracks corresponding to soprano (S), alto (A), tenor (T), and bass (B) with an end-to-end framework that integrates voice assignment through architectural design, pitch-range priors, order-continuity, and union-level supervision. We further introduce the voice assignment (VA) rate, a metric that quantifies the gap between part-aware and part-agnostic transcription performances.¹ Experimental results on the YouChorale dataset show that our model significantly outperforms the current state-of-the-art method [1] in part-agnostic transcription, with a 61.18% (0.237 vs. 0.382) relative improvement in note-level F1, and achieved an average F1 of 0.225 in multi-track part-aware choral transcription, serving as a baseline for upcoming research in this field. This work establishes the first end-to-end framework for part-aware choral transcription, where the code and model checkpoints can be accessed at: [redacted for anonymity.]

1. INTRODUCTION

Automatic music transcription (AMT) aims to convert an audio signal into a symbolic representation, such as a piano roll, MIDI file, or notated score [2]. AMT has achieved rapid advances in polyphonic single-instrument music [3–5], multi-instrument ensembles [6–11], and singing voice transcription [12–14]. Among these, piano transcription has seen particularly strong progress, supported by mature note-level models and large-scale paired audio–MIDI datasets [15, 16]. In particular, onset-conditioned and high-resolution piano transcription frameworks have substantially improved note-event estimation [3, 4].

¹In this paper, we use *part-aware* and *part-agnostic* to refer to multi-track and single-track choral transcription, respectively.

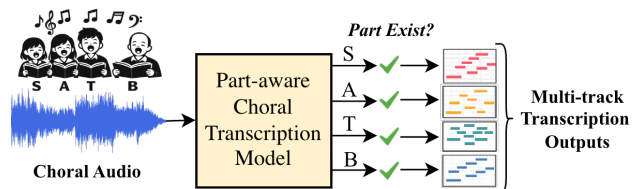


Fig. 1. Illustration of part-aware choral transcription: predicting active SATB parts and generating separate note-level MIDI tracks from a mixed choral recording.

Despite these advances, choral singing remains relatively underexplored in AMT [17]. Choral music commonly follows the four-part SATB setting: soprano (S), alto (A), tenor (T), and bass (B), where a subset or all four parts form simultaneous vocal streams [17–20]. Their overlapping ranges, soft onsets, smooth pitch transitions, and similar timbres make mixed choir recordings difficult to disentangle using cues effective for piano or multi-instrument transcription [21, 22]. Earlier choral-related MIR studies focused mainly on frame-level subtasks, including multi-F0 estimation [23], F0 streaming [24], source separation [21, 22], and pitch-level voice assignment from pitch salience representations [25, 26]. However, such frame-level pitch representations remain difficult to convert into note-level symbolic transcriptions, as they do not explicitly encode discrete note boundaries and therefore require additional decoding to recover note events. More recently, Yu and Duan introduced a Transformer-based framework for note-level transcription from mixed choral recordings [1]. However, its output does not detect and recover separate part tracks among SATB, which are required for score reconstruction [27, 28]. To the best of our knowledge, no existing work is found to produce part-aware, multi-track results for choral transcription.

To address this research gap, we formulate part-aware choral transcription as the task of first detecting the presence of a voice part among SATB from mixed choral recording, and then transcribing into four or fewer note-level MIDI tracks as illustrated in Fig. 1. This task is inherently challenging because the target parts are acoustically similar and cannot be easily distinguished by timbre or instrument class, which are key cues in many multi-track AMT tasks [6, 7, 29].

Based on this formulation, we propose PawCT (**P**art-**a**ware **C**horal **T**ranscription), an end-to-end framework for this task. Inspired by the existing high-resolution piano transcription architecture [4], PawCT introduces a voice assignment (VA) module at the architectural level that jointly predicts note-level activity for each part and SATB membership in an end-to-end manner. It also incorporates annotation-level voice assignment and involves a union-loss objective during training. Overall, PawCT is the first of its kind for part-aware choral transcription, a task we formulate and address in this paper. The main contributions of this paper are as follows:

- We define part-aware choral transcription as a note-level SATB transcription task from mixed choral recordings, and introduce a voice-assignment retention rate (VA Rate), computed as the ratio between average F1 across all parts and the overall part-agnostic F1, to quantify how much transcription quality is retained with the voice part assignment.
- We propose PawCT, an end-to-end framework that detects the active voice parts and transcribes them into multiple tracks from choral recordings. It further combines annotation-level voice-assignment priors based on pitch range and ordered continuity (Section 2.2) and supervision on the union of all parts’ outputs (Section 2.3) to improve the transcription performance.
- Results on the YouChorale dataset show that our model significantly outperform the existing state-of-the-art [1] for part-agnostic choral transcription, yielding a relative improvement of 61.18% (0.237 vs. 0.382) for note F1 and achieved an average F1 of 0.225 for part-aware choral transcription, serving as a baseline for upcoming research in this field.

2. METHOD

Fig. 2 summarises the proposed PawCT framework for part-aware choral transcription. The PawCT model performs multi-track choral transcription by mapping a shared CRNN representation to SATB-specific onset, offset, and frame predictions, followed by part-presence gating. To construct musically informed SATB targets, we further introduce annotation-level voice-assignment priors explained in Section 2.2. We also evaluate a part-agnostic variant of the proposed model, named **P**art-**a**gnostic **C**horal **T**ranscription (PagCT), which performs global transcription without SATB-specific prediction heads, and a two-stage Post-VA baseline that assigns SATB labels to notes after global transcription as explained in Section 2.4.

2.1. Part-aware Choral Transcription Architecture

The full PawCT architecture in Fig. 2(a) follows the high-resolution transcription framework of Kong et al. [4]. The

input audio is converted into a log-Mel spectrogram and encoded by a shared CRNN encoder. The encoder contains four convolutional blocks, each consisting of a 3×3 convolution, batch normalisation, Rectified Linear Unit (ReLU) activation, and average pooling along the frequency dimension. The resulting feature sequence is then passed through a projection layer and a two-layer bidirectional Gated Recurrent Unit (BiGRU) with hidden dimension of 256, producing a frame-aligned representation $\mathbf{h} \in \mathbb{R}^{B \times T \times 512}$, where B is the batch size and T is the number of frames.

The VA module maps this shared representation to four SATB streams. For each voice $v \in \{S, A, T, B\}$, separate linear heads project 512 into 88 pitch classes with sigmoid activations predict onset, offset, and frame probabilities over pitches. Since onset and offset labels are sparse event targets, we use the preliminary frame predictions as denser temporal context for frame refinement [30]. Specifically, the preliminary voice-wise frame outputs are concatenated over the voice and pitch dimensions and passed through an additional BiGRU with 256 hidden units per direction. A fully connected layer then projects the output back to $V \times P$ dimensions, where $V = 4$ and $P = 88$, and the result is reshaped to obtain refined voice-wise frame probabilities.

A part-presence head is applied in parallel to suppress inactive voices. It mean-pools $\mathbf{h}$ over time and passes the pooled representation through a two-layer MLP with output dimensions 256 and 4, producing a four-dimensional SATB presence vector. The predicted presence probabilities gate the voice-wise onset, offset, and frame outputs, so tracks predicted as inactive are suppressed and yield empty MIDI outputs.

2.2. Annotation-level Voice Assignment

Before constructing the training targets, the raw note annotations are mapped to the stream among canonical SATB. When the source part labels are reliable, labels such as S1/S2 or T1/T2 are merged into S or T. To handle cases in which part labels are ambiguous and impose stronger musical structure, we further apply rule-based voice-assignment priors during target construction, including the pitch range prior and ordered continuity. These priors affect only the supervision labels and are not used in the neural forward pass and inference stage.

2.2.1. Pitch Range Prior

The Pitch Range Prior (RP) assigns each note independently to the SATB voice with the lowest pitch compatibility cost. For a note event e with MIDI pitch p_e and candidate voice v , we define

$$\begin{aligned} \mathcal{C}_{\text{RP}}(e, v) = & \max(0, l_v - p_e) + \max(0, p_e - u_v) \\ & + 0.25 \frac{|p_e - m_v|}{12} + \lambda_{\text{part}} \mathbf{1}(v \neq v_{\text{part}}). \end{aligned} \quad (1)$$

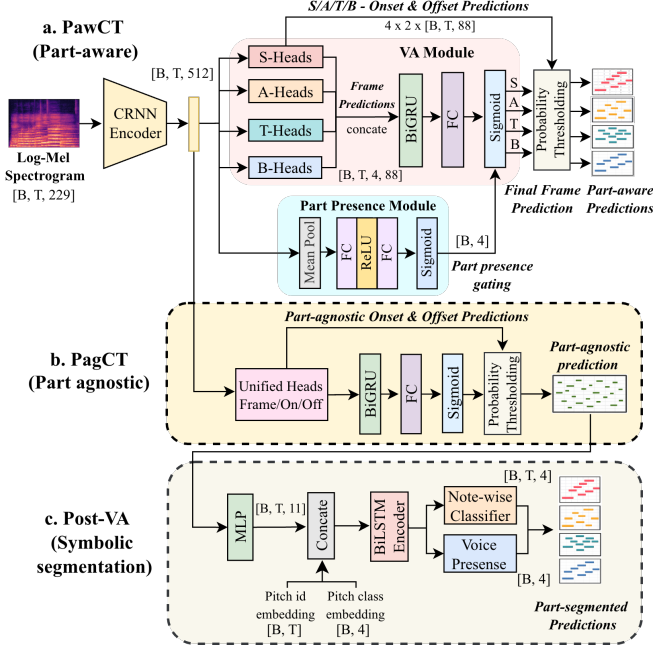


Fig. 2. Overview of the proposed frameworks: (a) PawCT performs part-aware transcription with SATB-specific prediction heads and part-presence gating. (b) PagCT is the part-agnostic variant of PawCT with unified onset, offset, and frame heads. (c) Post-VA assigns PagCT predicted notes to SATB parts using a symbolic segmentation assigner.

where $[l_v, u_v]$ is the expected pitch range and $m_v = (l_v + u_v)/2$ is the range center. The first two terms penalize pitches outside the candidate voice range in semitone units. The third term adds a weak preference for pitches near the center of the range: the distance is normalized by 12 semitones (i.e., one octave), and weighted by 0.25 so that it mainly acts as a tie-breaker within overlapping SATB ranges. The last term is used only when an original part label v_{part} is available, discouraging assignments that contradict the source annotation, and $1(\cdot)$ is the indicator function. We use the canonical MIDI ranges $S : [60, 88]$, $A : [55, 79]$, $T : [48, 72]$, and $B : [40, 67]$ that applied in previous choral voice assignment studies [31, 32]. The assigned voice is

$$\hat{v}_e = \arg \min_{v \in \{S, A, T, B\}} \mathcal{C}_{\text{RP}}(e, v). \quad (2)$$

This prior provides simple register-aware SATB labels but does not model temporal voice continuity.

2.2.2. Ordered Continuity

Ordered continuity (OC) extends RP by processing note events sequentially in onset synchronous groups rather than assigning each note independently. Notes are grouped by rounded onset frame and sorted by descending pitch within each group. For a candidate assignment of event

$e = (p_e, t_e^{\text{on}}, t_e^{\text{off}})$ to voice v , we define

$$\mathcal{C}_{\text{OC}}(e, v) = \mathcal{C}_{\text{RP}}(e, v) + \lambda_{\text{cont}} \frac{|p_e - \tilde{p}_v|}{12} + \lambda_{\text{ov}} \mathbf{1}(t_e^{\text{on}} < \tilde{t}_v^{\text{off}}). \quad (3)$$

where $\tilde{p}_v$ and $\tilde{t}_v^{\text{off}}$ denote the pitch and offset time of the most recent event assigned to voice v . If no such event exists, the continuity and overlap terms are omitted. The continuity term encourages smooth pitch motion within a voice, while the overlap term penalizes assigning a new note before the preceding note in the same voice has ended.

For each onset group G_t with $|G_t| \leq 4$, we search over order-preserving assignments to distinct SATB voices and choose

$$A_t^* = \arg \min_{A_t \in \mathcal{O}(G_t)} \sum_{e \in G_t} \mathcal{C}_{\text{OC}}(e, A_t(e)), \quad (4)$$

where $A_t \in \mathcal{O}(G_t)$ denotes a candidate assignment for onset group G_t , and $\mathcal{O}(G_t)$ denotes the set of one-to-one assignments that preserve the pitch ordering within the group. Among these candidates, the minimum cost assignment is denoted by A_t^* . After solving the assignment for group G_t , we update $\tilde{p}_v$ and $\tilde{t}_v^{\text{off}}$ and then move to the next group. In the rare case where more than four notes occur in the same onset group, the implementation instead uses independent minimum cost assignment.

2.3. PawCT Training Objective

We train PawCT with a multi-task objective that consists of voice-wise transcription loss, voice-presence loss, and a proposed union-level transcription loss to improve the consistency of the overall prediction in part-aware transcription. Let $\mathcal{A} = \{\text{fr}, \text{on}, \text{off}\}$ denote frame, onset, and offset targets. For each activity type $a \in \mathcal{A}$, the model predicts part-wise piano rolls $\hat{\mathbf{Y}}^a \in [0, 1]^{B \times T \times V \times P}$, where $V = 4$ for SATB voices and $P = 88$ for MIDI pitches. All transcription terms use masked binary cross-entropy (BCE):

$$\mathcal{B}(\hat{\mathbf{Y}}, \mathbf{Y}, \mathbf{M}) = \frac{\sum \mathbf{M} \ell_{\text{BCE}}(\hat{\mathbf{Y}}, \mathbf{Y})}{\max(1, \sum \mathbf{M})}, \quad (5)$$

where $\mathbf{M}$ masks out-of-segment frames. The voice-wise loss directly supervises the SATB-specific outputs, which can be written as

$$\mathcal{L}_{\text{voice}} = \sum_{a \in \mathcal{A}} \lambda_a^v \mathcal{B}(\hat{\mathbf{Y}}^a, \mathbf{Y}^a, \mathbf{M}^a). \quad (6)$$

Then, the voice presence head is trained with BCE loss over the segment level SATB presence labels,

$$\mathcal{L}_{\text{pres}} = \ell_{\text{BCE}}(\mathbf{z}^p, \mathbf{y}^p), \quad (7)$$

where $\mathbf{z}^p$ denotes the presence logits and $\mathbf{y}^p$ indicates which parts are active in the segment.

In addition, to preserve consistency with conventional global transcription, we also supervise the union outputs of all predicted SATB tracks, which are obtained by max pooling over voices, $\hat{\mathbf{Y}}^{a,\cup} = \max_v \hat{\mathbf{Y}}_v^a$, with union targets defined analogously. The proposed union loss is written as

$$\mathcal{L}_{\text{union}} = \sum_{a \in \mathcal{A}} \lambda_a^{\cup} \mathcal{B}(\hat{\mathbf{Y}}^{a,\cup}, \mathbf{Y}^{a,\cup}, \mathbf{M}^a). \quad (8)$$

Overall, the final objective is

$$\mathcal{L} = \mathcal{L}_{\text{voice}} + \lambda_u \mathcal{L}_{\text{union}} + \lambda_p \mathcal{L}_{\text{pres}}. \quad (9)$$

For ablations without union loss, we set $\lambda_u = 0$.

2.4. Part-agnostic Choral Transcription and Post-VA

We evaluate two variants to separate global transcription from vocal-part assignment. The part-agnostic model, denoted as PagCT in Fig. 2(b), removes the part-presence module and replaces the SATB-specific VA heads with a single global transcription head. It is trained with the BCE losses for onset, offset, and frame estimation defined in Eq. 5.

We also implement a two-stage Post-VA baseline to examine whether SATB assignment can be recovered after transcription. Unlike frame-level multiple-F0 voice-assignment methods [26], which transferred poorly to symbolic MIDI outputs in our preliminary experiments, we follow [33] and formulate Post-VA as note-level symbolic assignment. As shown in Fig. 2(c), each note is represented by 64-dimensional pitch and 16-dimensional pitch-class embeddings, concatenated with an 11-dimensional continuous feature vector. The combined representation is projected to 192 dimensions and encoded by a 2-layer bidirectional LSTM with 256 hidden units per direction. A four-way linear classifier predicts the SATB label for each note, while an auxiliary segment-level presence head, following the PawCT part-presence module, predicts active voices. The part-agnostic notes are then separated into four MIDI tracks. The model is trained using a masked per-note SATB cross-entropy loss combined with the segment-level voice-presence BCE loss defined in Eq. 7.

2.5. Note Event Decoding

At inference time, both PawCT and PagCT output probabilistic maps rather than discrete note events. Following [4], we detect local onset and offset peaks from the regression rolls, refine their positions using sub-frame shifts, and pair detected onsets with frame activity and offset cues to form MIDI notes. The same decoding procedure is used for both part-aware outputs among SATB and part-agnostic global outputs.

3. EXPERIMENTAL SETUPS

3.1. Data

We use YouChorale [1] as the main dataset, which contains raw audio and the notes with part labels. The training split is used for model training, while the validation and test splits are used for model selection and the main evaluation, respectively. YouChorale contains 22 hours and 25 minutes of a cappella choral recordings, comprising 452 recordings from 261 songs with manually aligned MIDI annotations. Most recordings follow the SATB setting: 418 contain all four parts (92.48%), while the remaining 34 contain only a subset of the SATB parts. To train the Post-VA model, we use the ground truth merged MIDI notes from the YouChorale training set together with their corresponding part labels.

We apply offline pitch transposition augmentation to the YouChorale training split. For each training example, the mixture audio, part-wise MIDI files, and note-level annotations are shifted by the same value in $\{-3, -2, -1, +1, +2, +3\}$ semitones. Together with the original training data, this expands the training set by a factor of seven, yielding approximately 129 hours and 38 minutes of audio.

3.2. Model Configurations

We evaluate the proposed framework under both part-agnostic and part-aware settings introduced in Section 2.

For part-agnostic transcription, we compare PagCT introduced in Section 2.4 against Onsets and Frames [3], MT3 [6], YourMT3+ [7], and the choral transcription model of Yu et al. [1]. Since [1] reports several configurations, we rerun them under the same YouChorale split, select the best configuration on the validation set, and report the corresponding test performance. PagCT is evaluated both with and without pitch-transposition augmentation, following the augmentation protocol of [1]. For part-aware transcription, we compare PawCT against two adapted baselines: YourMT3+ as a representative multitrack AMT model, and a modified part-aware variant of Yu et al. with four SATB-specific prediction branches. We further conduct ablation studies on PawCT with and without union loss, and with the annotation-level voice-assignment priors introduced in Section 2.2, denoted PawCT-RP and PawCT-OC. All part-aware models are trained on the augmented YouChorale training set with inference thresholds of 0.03, 0.01, and 0.005 for the frame, onset, and offset predictions, respectively.

We additionally evaluate the Post-VA pipeline as illustrated in Section 2.4 applied to part-agnostic outputs of the proposed PagCT, which assesses whether end-to-end joint modelling outperforms post-hoc symbolic voice assignment.

3.3. Training Settings

We follow the same preprocessing as [1]: all audio is down-sampled to 16 kHz, and log-mel spectrograms are computed following [4] with $N_{\text{FFT}} = 2048$, a hop size of 160 samples, and $M = 229$ mel filters. PawCT and PagCT training is performed with the Adam optimiser using a batch size of 8 and an initial learning rate of 10^{-4} . We train for 200k iterations and apply a multiplicative learning rate decay of 0.9 every 10k iterations. Early stopping is applied based on the validation loss with a patience of 20k, and the model checkpoint with the lowest validation loss is used for testing. For annotation-level voice assignment, we set the RP part-mismatch penalty to $\lambda_{\text{part}} = 2$, and the OC continuity and overlap penalties to $\lambda_{\text{cont}} = 0.35$ and $\lambda_{\text{ov}} = 4$, respectively. In the training objective, the voice-wise frame, onset, and offset losses are weighted by 1, 2, and 1, respectively, and the voice-presence loss is weighted by $\lambda_{\text{pres}} = 0.2$. When union-level supervision is used, the corresponding union-level frame, onset, and offset losses are all weighted by 1. For the Post-VA system, we train for up to 80 epochs with a learning rate of 10^{-4} and a batch size of 32, with early stopping based on the validation loss and 10 epochs patience. All training is performed on a single RTX 5090 GPU.

4. RESULTS AND DISCUSSION

4.1. Evaluation Metrics

For part-agnostic transcription, we report frame-level precision, recall, and F1, together with onset-based note precision, recall, and F1 under 50 ms and 100 ms onset tolerances, following [1]. For part-aware transcription, we report frame-level and note-level F1 with a 50 ms onset tolerance for each SATB part. We focus on F1 because it balances missed notes and false activations. We also report the average F1 across all voice parts. Finally, we define the voice assignment rate (VA Rate) as the ratio between the average part-aware F1 and the F1 of the corresponding part-agnostic model: $\text{VA Rate} = (\text{F1}_{\text{average}} / \text{F1}_{\text{part-agnostic}}) \times 100\%$. A higher VA Rate indicates that more transcription accuracy is preserved after voice assignment.

4.2. Part-agnostic Transcription

Table 1 reports the part-agnostic single-track transcription results on the YouChorale test set. In this setting, PagCT uses a single set of frame, onset, and offset heads and is compared with representative AMT baselines under the same evaluation protocol. Using F1 as the primary metric, PagCT outperforms both generic AMT baselines and the current state-of-the-art choral transcription model by Yu et al. [1]. Even without data augmentation, PagCT improves the 50ms onset F1 from 0.237 to 0.333. With pitch-transposition augmentation, the

note-level F1 further increases to 0.382, corresponding to a 61.18% relative improvement over Yu et al. [1].

These gains suggest that the CRNN-GRU encoder and sub-frame onset/offset regression are effective for modelling soft choral onsets, smooth pitch transitions, and note boundaries. Pitch transposition augmentation further exposes the model to a broader SATB register distribution, improving its robustness in note-event detection. We therefore use the augmented PagCT model as the part-agnostic reference for the subsequent VA Rate analysis.

4.3. Part-aware Transcription

Before analysing the transcription results, we note that the auxiliary assignment components are not the main bottleneck in isolation: the segment-level part-presence module achieves 100% F1, and the Post-VA symbolic segmenter reaches about 90% average F1 on segmented symbolic notes.

Table 2 reports part-aware transcription results on YouChorale. Directly adapting generic multitrack AMT is insufficient for note-level part-aware transcription. Although the modified YourMT3+ baseline obtains a relatively high note-level VA Rate of 62.20%, its average note F1 is only 0.079, showing that VA Rate must be interpreted together with absolute transcription quality. The choral-specific Yu et al. [1] adaptation performs better, reaching 0.165 average note F1, but remains below all end-to-end PawCT variants. PawCT-OC further improves this score to 0.225, a 36.36% relative improvement over the adapted choral baseline.

Union-level supervision is important for stable multitrack learning. Removing the union loss yields 0.379 average frame F1 and 0.190 average note F1, while adding it raises these scores to 0.458 and 0.217 and improves note F1 for every SATB part. This indicates that the union objective anchors the voice-wise heads to the global pitch content while still allowing part-specific specialisation.

The annotation-level VA priors have different effects at the frame and note levels. The pitch range prior improves average frame F1 from 0.458 to 0.510 and frame-level VA Rate from 62.06% to 69.11%, suggesting that register constraints suppress implausible frame-level activations. However, its average note F1 drops from 0.217 to 0.209, indicating that pitch range alone is insufficient in overlapping SATB registers. Adding ordered continuity addresses this limitation: PawCT-OC achieves the best average note F1 among end-to-end models, 0.225, and the highest note-level VA Rate among PawCT variants, 58.90%.

Finally, the two-stage Post-VA result highlights the limitation of symbolic reassignment after transcription. Post-VA + PagCT obtains a competitive average frame F1 of 0.489, but only 0.175 average note F1 and 45.81% note-level VA Rate. Thus, although the global transcriber preserves substantial frame-level activity, symbolic reassignment cannot reliably recover SATB labels when predicted notes are missed,

Model	Frame			Onset (50ms)			Onset (100ms)		
	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1
Onsets and Frames [3]	0.806	0.326	0.428	0.450	0.178	0.242	0.688	0.248	0.344
MT3 [6]	0.590	0.243	0.344	0.117	0.148	0.127	0.200	0.255	0.217
YourMT3+ [7]	0.529	0.423	0.457	0.145	0.119	0.127	0.252	0.207	0.221
Yu et al. [1]	0.678	0.619	0.647	0.251	0.232	0.237	0.366	0.341	0.347
PagCT w/o Aug.	<u>0.690</u>	<u>0.833</u>	0.751	0.346	0.329	0.333	0.539	0.506	0.516
PagCT (Proposed)	0.626	0.909	<u>0.738</u>	<u>0.398</u>	0.378	0.382	<u>0.597</u>	0.569	0.573

Table 1. Part-agnostic single-track transcription results on the YouChorale test set. “w/o Aug.” denotes training without data augmentation, as described in Section 3.1.

Model	Soprano		Alto		Tenor		Bass		Average		VA Rate (%)	
	Frame	Note	Frame	Note	Frame	Note	Frame	Note	Frame	Note	Frame	Note
<i>End-to-end Settings</i>												
*YourMT3+ [7]	0.371	0.083	0.297	0.068	0.294	0.073	0.371	0.093	0.333	0.079	72.87	62.20
*Yu et al. [1]	0.528	0.183	<u>0.451</u>	0.151	0.436	0.138	0.560	0.186	0.494	0.165	76.35	69.62
PawCT w/o union-loss	0.417	0.192	0.338	0.169	0.343	0.178	0.419	0.222	0.379	0.190	51.36	49.74
PawCT	0.530	<u>0.228</u>	0.430	<u>0.191</u>	0.429	<u>0.197</u>	0.552	<u>0.252</u>	0.458	<u>0.217</u>	62.06	56.81
PawCT-RP	0.542	0.210	0.453	0.185	0.458	0.194	0.585	0.247	0.510	0.209	69.11	54.71
PawCT-OC (Proposed)	<u>0.535</u>	0.231	0.447	0.197	<u>0.451</u>	0.208	<u>0.578</u>	0.264	<u>0.503</u>	0.225	68.16	58.90
<i>Two-stage Settings</i>												
PagCT + Post-VA	0.501	0.180	0.439	0.145	0.447	0.156	0.568	0.219	0.489	0.175	66.26	45.81

Table 2. Part-aware frame-level and note-level transcription results on the YouChorale test set. Note-level results use Onset F1 with a 50 ms tolerance. All models are trained on YouChorale with data augmentation. The VA Rate is a retention metric computed as $VA\ Rate = (F1_{average}/F1_{part-agnostic}) \times 100\%$. * denotes models adapted for part-aware transcription. “w/o union-loss” removes the proposed union loss in Section 2.3. PawCT-RP uses pitch range prior annotation-level VA, while PawCT-OC further applies order-continuity VA on top of RP, as described in Section 2.2.

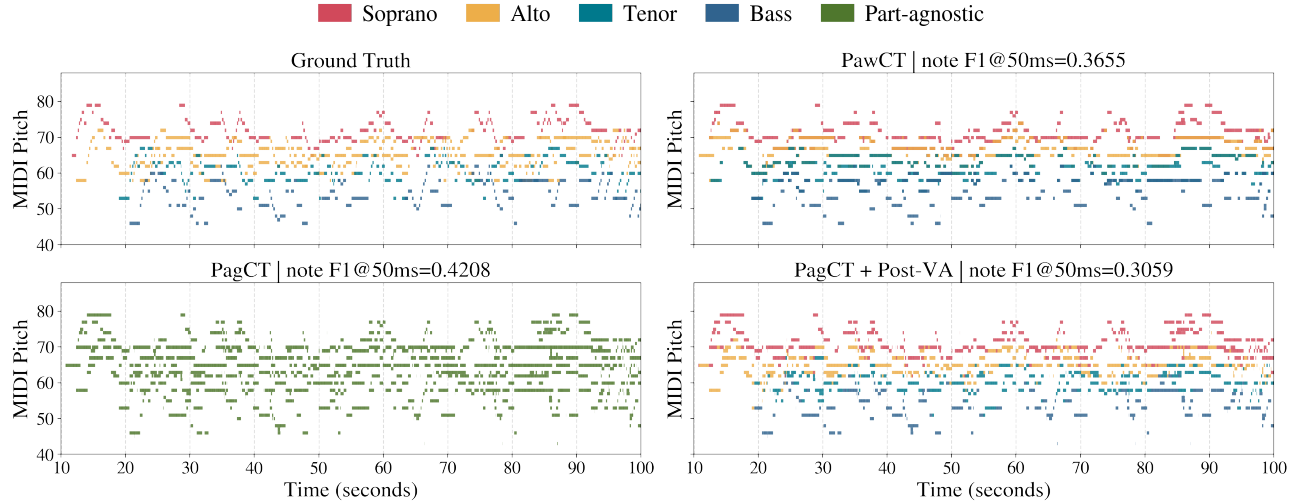


Fig. 3. Piano-roll visualisation of transcription results for “Exsultate Deo” by Palestrina (ID: jd2_r4PK5dc) from the You-Chorale test set. The figure compares the ground truth, PagCT, PawCT-OC, and Post-VA + PagCT.

merged, over-extended, or crossing in pitch. Compared with Post-VA + PagCT, PawCT-OC improves average note F1 by 0.050 absolute, or 28.57% relative, and raises note-level VA Rate by 13.09 percentage points. These results support joint acoustic transcription and voice assignment over post-hoc symbolic reassignment.

4.4. Visualisation Analysis

Fig. 3 illustrates the main error patterns behind the quantitative results in Table 1 and 2. PagCT preserves many note events in the global transcription, but it often over-extends note durations, producing tails beyond the ground-truth offsets. Post-VA + PagCT inherits these event-boundary errors and further introduces voice-label inconsistencies when SATB parts overlap or cross in pitch, since the symbolic reassignment stage no longer has access to the acoustic evidence in the original mixture.

By contrast, PawCT-OC produces cleaner part-wise tracks with fewer visibly over-extended notes and more coherent SATB trajectories. It is also more conservative, missing some short or transient notes, but the predicted notes are more consistently assigned to the correct vocal parts. This qualitative pattern matches the quantitative trend: Post-VA + PagCT retains competitive frame-level activity but underperforms at the note level, whereas PawCT-OC achieves the strongest average note F1 among the end-to-end part-aware models. The visualisation therefore supports the central claim that note-level part-aware transcription benefits from joint acoustic transcription and voice assignment, rather than symbolic reassignment after global transcription.

5. CONCLUSION AND FUTURE WORK

This paper introduced part-aware choral transcription, a note-level task that estimates active vocal parts and transcribes the corresponding note events from choral recordings. To address this task, we proposed PawCT, an end-to-end framework combining voice-specific acoustic prediction, part-presence gating, union-level supervision, and annotation-level voice-assignment priors. Experiments on YouChorale show that PagCT, the part-agnostic counterpart of PawCT for part-agnostic global transcription, improves part-agnostic note-level transcription over the current choral AMT baseline, while PawCT-OC achieves the strongest note-level part-aware performance among the evaluated variants. Ablation results further show that union-level supervision stabilises multitask learning, ordered-continuity targets improve SATB part assignment, and joint acoustic modelling outperforms Post-VA symbolic reassignment. Future work will address overlapping inner voices, ambiguous note boundaries, and short transient notes through duration-aware objectives, voice-leading or harmony-constrained decoding, and pretrained audio representations.

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